a grid-based puzzle game

COLOR SHIFT GRID

Drăghicescu Roxana-Petrina

Szenteș Iulia

PROJECT SPECIFICATIONS

Purpose of the Application

- interactive grid-based puzzle game
- develops logical thinking and strategic planning
- each move affects multiple elements
- creates chain reactions across the grid

Functional Requirements

The system must provide:

- 2D grid (e.g. 5x5) with colored cells
- user interaction through cell clicks
- color transformation (cell + neighbors)
- game modes: classic, challenge and pattern
- undo and restart functionality

Game Rules

- each cell stores a color as an integer (0-3)
- colors follow a cyclic order:

Red – Blue – Green – Yellow – Red

- a move updates: the selected cell and the direct neighbors (N, S, E, W)
- update rule:

new = (old+1) mod 4

- win condition depends on the selected game mode

Game modes

- Classic Mode
  - no move limit
  - undo
  - restart
  - progress
- Challenge Mode
  - move limit
  - limited undo
  - restart
  - progress
- Pattern Mode
  - target pattern
  - undo
  - restart
  - progress

Level Generation

- start from a solved grid
- apply a sequence of valid moves
- resulting grid becomes the level
- ensures every level is solvable

Constraints

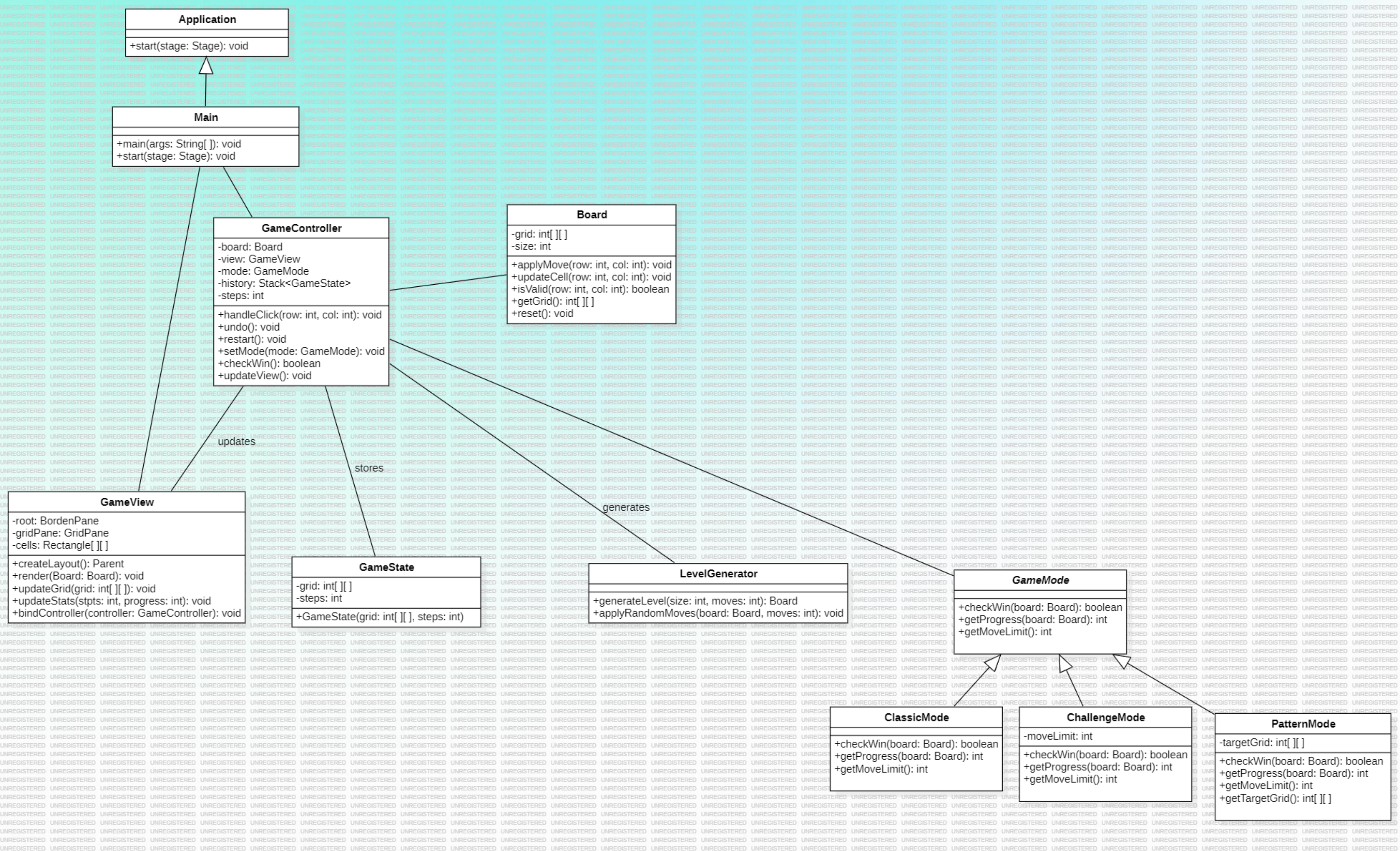
- grid size is fixed (e.g. 5x5)
- only valid positions are affected by moves
- game rules are deterministic
- constraints vary depending on game mode

UML OVERVIEW

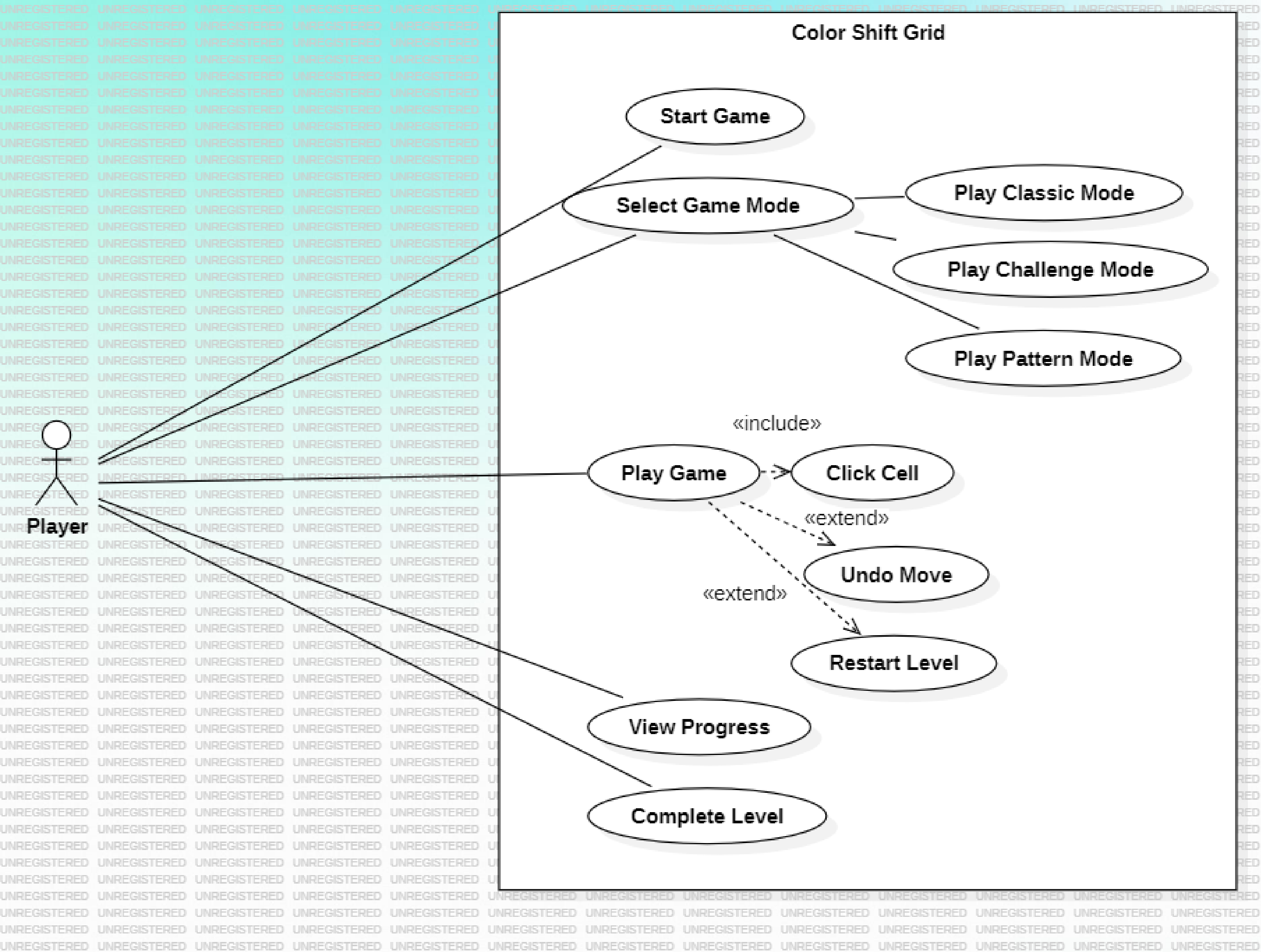
System Modeling (UML)

- Class Diagram
- Use Case Diagram
- State Diagram

CLASS DIAGRAM



USE CASE DIAGRAM



STATE DIAGRAM